

FRANCESCO GRUSSU

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EDUCATION AND TRAINING

PhD in MR Physics	2012-2016	University College London (UCL), UK Mark: viva passed with no corrections. Award date: 28/03/2016. Thesis: " <i>Microstructural imaging of the human spinal cord with advanced diffusion MRI</i> "
Master's Degree in Bioengineering	2010-2012	University of Genoa, Italy Mark: 110 out of 110 <i>cum laude</i> & <i>Dignità di Stampa</i> (Examination Panel award). Dissertation: " <i>A study on a bidirectional brain-machine interface inspired by the corticospinal control of movement</i> ".
Bachelor's Degree in Biomedical Engineering	2006-2009	University of Cagliari, Italy Mark: 110 out of 110 <i>cum laude</i> . Dissertation: " <i>Real time wavelet denoising on a DSP of neural signals coming from the peripheral nervous system</i> ".

WORK EXPERIENCE

Vall d'Hebron Institute of Oncology	Jan.2025-now	Senior investigator, VHIO — SPAIN Affiliations: Radiomics Group, Biomedical Research Department. Role: supervisor of research line in diffusion MRI. Design of MRI studies.
Vall d'Hebron Institute of Oncology	Oct.2020-Dec.24	Senior post-doc, VHIO — BARCELONA, SPAIN Affiliations: Radiomics Group, Clinical Research Department. Role: development of microstructural MRI techniques in cancer.
University College London	Sep.2020-now	Honorary Senior Fellow, UCL — LONDON, UK Affiliations: Queen Square Institute of Neurology. Role: collaborator in MRI development in multiple sclerosis.
University College London	Feb.2016-Sept.20	Research Associate, UCL — LONDON, UK Affiliations: Institute of Neurology; Centre for Medical Image Computing. Responsibilities: quantitative MRI development (spinal cord, brain, prostate).
	May-June 2012	Research Assistant, QMUL — LONDON, UK

Queen Mary Univ.
of London

Affiliation: School of Electronic Engineering and Computer Science (EECS).
Responsibilities: image segmentation (Computed Tomography Angiography).

RESEARCH VISITS

Visiting Researcher	Feb.-March 2024	Champalimaud Foundation, Portugal
		Affiliation: Preclinical MRI lab, Champalimaud Foundation, Lisbon, Portugal. Training: preclinical MRI and multi-echo gradient echo imaging.
Visiting Researcher	Sept.-Nov. 2017	New York University (NYU), USA
		Affiliation: Radiology, Langone Medical Center, New York City (USA). Training: advanced denoising techniques for diffusion MRI.

IMPACT STATS ON THE 13/05/2026

<i>h-index</i>	Google Scholar h-index: 25. Web of Science h-index: 20.
<i>Citations</i>	Total Google Scholar citations: 2689. Web of Science citations: 1600.

FUNDING: PROJECT GRANTS

EU4Health 2026	EU4Health project grant , call EU4H-2026-SANTE-PJ-o8, European Commission. “Scale-up biotech AI innovation by leveraging responsible data use under the European Health Data Space” (SCAILED). Award: 92,362.4 €. Role: co-investigator for VHIO. Principal investigator and primary coordinator: Gu Wei, Luxembourg National Data Service (LNDS). Consortium: 47 institutes/centres, with total budget of 14,384,759.99 €. Duration: 01/09/2026-31/08/2029 (grant agreement in preparation; awarded 21/04/2026).
CaixaResearch 2025	CaixaResearch Institute Innovation Hub 2025 . CaixaResearch Institute, Barcelona (Spain). “Computational Magnetic Resonance for metastatic prostate cancer”. Award: 30,000 €. Role: principal investigator. Duration: 01/01/2026-31/12/2026. Competitive call open to researchers from Innovation Hub institutions (VHIO; IrsiCaixa; ISGlobal; BarcelonaBeta).
AEI Generación de Conocimiento/ Knowledge Generation 2024	2024 Knowledge Generation Projects (Proyectos de Generación de Conocimiento) , National Research Agency (Agencia Estatal de Investigación (AEI)), Ministry of Science, Innovation and Universities, Spain. “Magnetic Resonance Imaging (MRI) foundational artificial intelligence for non-invasive, histologically-meaningful cancer characterisation (MRI-Found-Histo)”. Award: 118,750 €. Role: principal investigator. Duration: 01/09/2025-31/08/2028. Code: PID2024-158670OA-I00.
UCL pump-priming award 2017	Departmental seed grant . Centre for Medical Image Computing Pump-priming Award at University College London (UCL). “Enabling multi-site high precision spinal cord MRI”. Award: GBP 23,900. Role: principal investigator. Duration: 01/07/2017-30/06/2018.

FUNDING: FELLOWSHIPS AND STUDENTSHIPS

“la Caixa” Junior Leader Fellowship 2022	2022 Junior Leader Retaining post-doctoral fellowship , “la Caixa” Foundation, Spain. “New-generation oncological MRI (New-OncoMRI): development, validation and application”. Award: €294,900. Role: principal investigator and fellow. Duration: 30/09/2022-29/09/2025. Code: ID 100010434, fellowship number LCF/BQ/PR22/11920010.
Beatriu de Pinós Fellowship 2020	2020 Beatriu de Pinós post-doctoral fellowship , AGAUR, Secretary of Universities and Research (Govt of Catalonia, Spain). “Advancing Magnetic

Resonance Imaging against liver cancer". **Award:** €144,300. **Role:** principal investigator and fellow. **Duration:** 01/01/2022-31/12/2024, renounced on 29/09/2022 due to incompatibility with the "la Caixa" Junior Leader fellowship. **Code:** 2020 BP 00117.

UCL Grand
Challenge PhD
studentship 2012

UCL School of Life and Medical Science *Grand Challenge PhD Studentship*, awarded to work on the project "Axonal density as MR imaging biomarker: from bench to bedside". **Award:** 59,000 GBP. **Description:** PhD studentship covering a 3-year student stipend, tuition fees at UK/EU rate, and research costs. **Duration:** 09/2012-12/2015.

SCIENTIFIC ARTICLES AND REVIEWS — SENIOR AUTHORSHIP

Magn Reson Med
2026

"Simulation-informed evaluation of microvascular parameter mapping for diffusion MR imaging of solid tumours". Voronova AK et al, Grussu F. *Magnetic Resonance in Medicine* (2026), 96(1): 387-402, doi: [10.1002/mrm.70318](https://doi.org/10.1002/mrm.70318). Senior (last) author, corresponding author.

Comm Bio 2025

"Histology-informed microstructural diffusion simulations for MRI cancer characterisation — the Histo-microSim framework". Grigoriou A et al, Grussu F. *Communications Biology* (2025), 8: 1695, doi: [10.1038/s42003-025-09096-3](https://doi.org/10.1038/s42003-025-09096-3). Senior (last) author, corresponding author.

MedIA 2025

"SpinFlowSim: a blood flow simulation framework for histology-informed diffusion MRI microvasculature mapping in cancer". Voronova AK et al, Grussu F. *Medical Image Analysis* (2025), 102: 103531, doi: [10.1016/j.media.2025.103531](https://doi.org/10.1016/j.media.2025.103531). Senior (last) author, corresponding author.

JMRI 2025

"Enhancing tumor microstructural quantification with machine learning and diffusion-relaxation MRI ". Macarro C et al, Grussu F. *Journal of Magnetic Resonance Imaging* (2025), 61(2): 1018-1021, doi: [10.1002/jmri.29484](https://doi.org/10.1002/jmri.29484). Senior (last) author, corresponding author.

JMRI 2024

Review: "Advanced diffusion-weighted MRI for cancer microstructure assessment in body imaging, and its relationship with histology". Fokkinga E, Hernandez-Tamames JA, Ianus A, Nilsson M, Tax CMW, Perez-Lopez R, Grussu F. *Journal of Magnetic Resonance Imaging* (2024), 60(4): 1278-1304, doi: [10.1002/jmri.29144](https://doi.org/10.1002/jmri.29144). Senior (last) author, corresponding author.

Front Neurol 2021

"Comparison of neurite orientation dispersion and density imaging and two-compartment spherical mean technique parameter maps in multiple sclerosis". Johnson D et al, Grussu F. *Frontiers in Neurology* (2021), 12: 662855, doi: [10.3389/fneur.2021.662855](https://doi.org/10.3389/fneur.2021.662855). Senior (last) author.

SCIENTIFIC ARTICLES AND REVIEWS — (CO)-FIRST AUTHORSHIP

J Immunother
Cancer 2025

"Integrating C-reactive protein flare and early MRI dynamics for enhanced prediction of immunotherapy response". Navarro D*, Grussu F* et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e012143, doi: [10.1136/jitc-2025-012143](https://doi.org/10.1136/jitc-2025-012143). *: DN and FG are joint first authors (equal contribution).

Comm Med 2025

"Clinically feasible liver tumour cell size measurement through histology-informed in vivo diffusion MRI". Grussu F et al. *Communications Medicine* (2025), 5: 535, doi: [10.1038/s43856-025-01246-2](https://doi.org/10.1038/s43856-025-01246-2). Corresponding author.

Cell Rep Med 2024

"An accessible deep learning tool for voxel-wise classification of brain malignancies from perfusion MRI". Garcia-Ruiz A*, Pons-Escoda A*, Grussu F* et al. *Cell Reports Medicine* (2024), 5(3): 101464, doi: [10.1016/j.xcrm.2024.101464](https://doi.org/10.1016/j.xcrm.2024.101464). *: AGR, APE and FG are joint first authors (equal contribution).

Magn Reson Med
2022

"Diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time: Insights from simulations, 9.4T imaging, and histology". Grussu F et al. *Magnetic*

Resonance in Medicine (2022), 88(1): 365-379, doi: [10.1002/mrm.29174](https://doi.org/10.1002/mrm.29174).
Corresponding author.

- Front Phys* 2021 “Feasibility of data-driven, model-free quantitative MRI protocol design: application to brain and prostate diffusion-relaxation imaging”. Grussu F et al. *Frontiers in Physics* (2021), 9: 752208, doi: [10.3389/fphy.2021.752208](https://doi.org/10.3389/fphy.2021.752208). Corresponding author.
- Sem Ultrasound CT MRI* 2021 **Review:** “Diffusion-weighted imaging: recent advances and applications”. Martinez-Heras E*, Grussu F*, et al. *Seminars in Ultrasound, CT and MRI* (2021), 42(5): 490-506, doi: [10.1053/j.sult.2021.07.006](https://doi.org/10.1053/j.sult.2021.07.006). *: EMH and FG are joint first authors (equal contribution).
- NeuroImage* 2020 “Multi-parametric quantitative in vivo spinal cord MRI with unified signal readout and image denoising”. Grussu F et al. *NeuroImage* (2020), 217: 116884, doi: [10.1016/j.neuroimage.2020.116884](https://doi.org/10.1016/j.neuroimage.2020.116884). Corresponding author.
- Magn Reson Med* 2019 “Relevance of time-dependence for clinically viable diffusion imaging of the spinal cord”. Grussu F et al. *Magnetic Resonance in Medicine* (2019), 81(2): 1247-1264, doi: [10.1002/mrm.27463](https://doi.org/10.1002/mrm.27463). Corresponding author.
- Ann Clin Transl Neurol* 2017 “Neurite dispersion: a new marker of multiple sclerosis spinal cord pathology?”. Grussu F, Schneider T et al. *Annals of Clinical and Translational Neurology* (2017), 4(9):663-679, doi: [10.1002/acn3.445](https://doi.org/10.1002/acn3.445). FG and TS are joint first authors (equal contribution). Paper featured in *Nature Reviews Neurology* “Research Highlights” (Patel M, *Nat Rev Neur* (2017), 13(10): 578, doi: [10.1038/nrneurol.2017.127](https://doi.org/10.1038/nrneurol.2017.127)).
- J Neurosci Meth* 2016 “A framework for optimal whole-sample histological quantification of neurite orientation dispersion in the human spinal cord”. Grussu F et al. *Journal of Neuroscience Methods* (2016), 273: 20-32, doi: [10.1016/j.jneumeth.2016.08.002](https://doi.org/10.1016/j.jneumeth.2016.08.002). Corresponding author.
- NeuroImage* 2015 “Neurite orientation dispersion and density imaging of the healthy cervical spinal cord in vivo”. Grussu F et al. *NeuroImage* (2015), 111: 590-601, doi: [10.1016/j.neuroimage.2015.01.045](https://doi.org/10.1016/j.neuroimage.2015.01.045). Corresponding author.

SCIENTIFIC ARTICLES AND REVIEWS — CO-AUTHORSHIP

- Eur Radiol* 2026 **Review:** “ESR Essentials: diffusion-weighted MRI—practice recommendations by the European Society for Magnetic Resonance in Medicine and Biology”. Palombo M, Bodini B, Grussu F et al. *European Radiology* (2026), 36: 2198-2208, doi: [10.1007/s00330-025-12033-x](https://doi.org/10.1007/s00330-025-12033-x).
- JMRI* 2026 “The sense of smell (SoS) atlas: its creation and first application to investigate COVID-19 related anosmia with a comprehensive quantitative MRI protocol”. Gaviraghi M, ..., Grussu F et al. *Journal of Magnetic Resonance Imaging* (2026), 63(2): 574-593, doi: [10.1002/jmri.70128](https://doi.org/10.1002/jmri.70128).
- Clin Cancer Res* 2026 “Broad utility of ultrasensitive analysis of ctDNA dynamics across solid tumors treated with immunotherapy”. Garralda E, ..., Grussu F et al. *Clinical Cancer Research* (2026), 32(2): 333-349, doi: [10.1158/1078-0432.CCR-25-2312](https://doi.org/10.1158/1078-0432.CCR-25-2312).
- Magn Res Med* 2025c **Review:** “Considerations and recommendations from the ISMRM Diffusion Study Group for preclinical diffusion MRI: Part 3—Ex vivo imaging: Data processing, comparisons with microscopy, and tractography”. Schilling KG, Howard AFD, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2561-2582, doi: [10.1002/mrm.30424](https://doi.org/10.1002/mrm.30424).
- Magn Res Med* 2025b **Review:** “Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 2—Ex vivo imaging: Added value and acquisition”. Schilling KG, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2535-2560, doi: [10.1002/mrm.30435](https://doi.org/10.1002/mrm.30435).
- Magn Res Med* 2025a **Review:** “Considerations and recommendations from the ISMRM diffusion study

- group for preclinical diffusion MRI: Part 1: *In vivo* small-animal imaging". Jelescu IO, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2507-2534, doi: [10.1002/mrm.30429](https://doi.org/10.1002/mrm.30429).
- Imag Neurosci* 2025 "Body size and intracranial volume interact with the structure of the central nervous system: A multi-center *in vivo* neuroimaging study". Labounek R, ... Grussu F et al. *Imaging Neuroscience* (2025), 3: 00559, doi: [10.1162/imag-a.00559](https://doi.org/10.1162/imag-a.00559).
- Sci Rep* 2025 "Evaluation of magnetic resonance spectroscopy total sodium concentration measures, and associations with microstructure and physical impairment in cervical myelopathy". Solanky B, ... Grussu F et al. *Scientific Reports* (2025), 15: 7014, doi: [10.1038/s41598-025-91658-w](https://doi.org/10.1038/s41598-025-91658-w).
- J Immunother Cancer* 2025 "Radiomics signature for dynamic monitoring of tumor inflamed microenvironment and immunotherapy response prediction". Bernatowicz K, ... Grussu F et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e009140, doi: [10.1136/jitc-2024-009140](https://doi.org/10.1136/jitc-2024-009140).
- NeuroImage Rep* 2024 "Investigating the relationship between thalamic iron concentration and disease severity in secondary progressive multiple sclerosis using quantitative susceptibility mapping: Cross-sectional analysis from the MS-STAT2 randomised controlled trial". Williams T, ..., Grussu F, ..., Chataway J; On behalf of The UCL MS-STAT2 investigators. *NeuroImage: Reports* (2024), 4(3): 100216, doi: [10.1016/j.ynrp.2024.100216](https://doi.org/10.1016/j.ynrp.2024.100216).
- Eur Urol* 2024 "Whole-body magnetic resonance imaging as a treatment response biomarker in castration-resistant prostate cancer with bone metastases: the iPROMET clinical trial". Garcia-Ruiz A, ... Grussu F et al. *European Urology* (2024), 86(3): 272-274, doi: [10.1016/j.eururo.2024.02.016](https://doi.org/10.1016/j.eururo.2024.02.016).
- Mult Scler* 2024 "What contributes to disability in progressive MS? A brain and cervical cord-matched quantitative MRI study". Tur C, ..., Grussu F, ..., Gandini Wheeler-Kingshott CAM. *Multiple Sclerosis Journal* (2024), e-pub ahead of print, doi: [10.1177/13524585241229969](https://doi.org/10.1177/13524585241229969).
- Radiology: AI* 2024 "Identification of precise 3D CT radiomics for habitat computation by machine learning in cancer". Prior O, ..., Grussu F, Bernatowicz K*, Perez-Lopez R*. *Radiology: Artificial Intelligence* (2024), 6(2): e230118, doi: [10.1148/ryai.230118](https://doi.org/10.1148/ryai.230118). *: KB and RPL are joint senior authors.
- Movement Disord* 2023 "Multimodal analysis of the visual pathways in Friedreich's Ataxia reveals novel biomarkers". Thomas-Black G, ... Grussu F et al. *Movement Disorders* (2023), 38(6): 959-969, doi: [10.1002/mds.29277](https://doi.org/10.1002/mds.29277).
- Sci Rep* 2023 "Feasibility of *in vivo* multi-parametric quantitative magnetic resonance imaging of the healthy sciatic nerve with a unified signal readout protocol". Boonsuth R, Battiston M, Grussu F et al. *Scientific Reports* (2023), 13: 6565, doi: [10.1038/s41598-023-33618-w](https://doi.org/10.1038/s41598-023-33618-w).
- Front Neuroinform* 2023 "Patterns of inflammation, microstructural alterations, and sodium accumulation define multiple sclerosis subtypes after 15 years from onset". Ricciardi A, Grussu F et al. *Frontiers in Neuroinformatics* (2023), 17: 1060511, doi: [10.3389/fninf.2023.1060511](https://doi.org/10.3389/fninf.2023.1060511).
- Neurology* 2023 "Differentiating Multiple Sclerosis from AQP4-Neuromyelitis Optica Spectrum Disorder and MOG-antibody disease with imaging". Cortese R, ... Grussu F et al. *Neurology* (2023), 100(3): e308-e323, doi: [10.1212/WNL.0000000000201465](https://doi.org/10.1212/WNL.0000000000201465).
- Am J Neuroradiol* 2022 "Diffuse large B-cell Epstein-Barr virus-positive primary CNS lymphoma in non-AIDS patients: high diagnostic accuracy of DSC perfusion metrics". Pons-Escoda A, ... Grussu F et al. *American Journal of Neuroradiology* (2022), 43(11): 1567-1574, doi: [10.3174/ajnr.A7668](https://doi.org/10.3174/ajnr.A7668).
- Magn Reson Med* 2022c "Multi-echo quantitative susceptibility mapping: how to combine echoes for accuracy and precision at 3 Tesla". Biondetti E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2101-2116, doi: [10.1002/mrm.29365](https://doi.org/10.1002/mrm.29365).

- Magn Reson Med* 2022b "SENSE EPI reconstruction with 2D phase error correction and channel-wise noise removal". Powell E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2157-2166, doi: [10.1002/mrm.29349](https://doi.org/10.1002/mrm.29349).
- Magn Reson Med* 2022a "Comparison of multicenter MRI protocols for visualizing the spinal cord gray matter". Cohen-Adad J, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(2): 849-859, doi: [10.1002/mrm.29249](https://doi.org/10.1002/mrm.29249).
- Eur Radiol* 2022 "Voxel-level analysis of normalized DSC-PWI time-intensity curves: a potential generalizable approach and its proof of concept in discriminating glioblastoma and metastasis". Pons-Escoda A, ..., Grussu F et al. *European Radiology* (2022), 32: 3705-3715, doi: [10.1007/s00330-021-08498-1](https://doi.org/10.1007/s00330-021-08498-1).
- BMJ Open* 2022 "Histo-MRI map study protocol: a prospective cohort study mapping MRI to histology for biomarker validation and prediction of prostate cancer". Singh S, ..., Grussu F et al. *BMJ Open* (2022), 12: e059847, doi: [10.1136/bmjopen-2021-059847](https://doi.org/10.1136/bmjopen-2021-059847).
- NeuroImage Clin* 2022 "Spatial patterns of brain lesions assessed through covariance estimations of lesional voxels in multiple sclerosis: the SPACE-MS technique". Tur C, Grussu F et al. *NeuroImage: Clinical* (2022), 33: 102904, doi: [10.1016/j.nicl.2021.102904](https://doi.org/10.1016/j.nicl.2021.102904).
- Front Neurol* 2021 "Assessing lumbar plexus and sciatic nerve damage in relapsing-remitting multiple sclerosis using magnetisation transfer ratio". Boonsuth R, ..., Grussu F et al. *Frontiers in Neurology* (2021), 12: 763143, doi: [10.3389/fneur.2021.763143](https://doi.org/10.3389/fneur.2021.763143).
- Sci Rep* 2021 "Robust imaging habitat computation using voxel-wise radiomics features". Bernatowicz K, Grussu F et al. *Scientific Reports* (2021), 11: 20133, doi: [10.1038/s41598-021-99701-2](https://doi.org/10.1038/s41598-021-99701-2).
- Sci Data* 2021 "Open-access quantitative MRI data of the spinal cord and reproducibility across participants, sites and manufacturers". Cohen-Adad J, ..., Grussu F et al. *Scientific Data* (2021), 8: 219, doi: [10.1038/s41597-021-00941-8](https://doi.org/10.1038/s41597-021-00941-8).
- Nat Protoc* 2021 "Generic acquisition protocol for quantitative MRI of the spinal cord". Cohen-Adad J, ..., Grussu F et al. *Nature Protocols* (2021), 16: 4611-4632, doi: [10.1038/s41596-021-00588-0](https://doi.org/10.1038/s41596-021-00588-0).
- Brain* 2021 "Brain microstructural and metabolic alterations detected in vivo at the onset of the first demyelinating event". Collorone S, ..., Grussu F et al. *Brain* (2021), 144: 1409-1421, doi: [10.1093/brain/awabo43](https://doi.org/10.1093/brain/awabo43).
- NeuroImage* 2021 "Uncertainty modelling in deep learning for safer neuroimage enhancement: demonstration in diffusion MRI". Tanno R, ... Grussu F et al. *NeuroImage* (2021), 225: 117366, doi: [10.1016/j.neuroimage.2020.117366](https://doi.org/10.1016/j.neuroimage.2020.117366).
- Mult Scler* 2020b "Reduced neurite density in the brain and cervical spinal cord in relapsing-remitting multiple sclerosis: A NODDI study". Collorone S, Cowley N, Grussu F et al. *Multiple Sclerosis Journal* (2020), 26(13): 1647-1657, doi: [10.1177/1352458519885107](https://doi.org/10.1177/1352458519885107).
- Mult Scler* 2020a "A multi-shell multi-tissue diffusion study of brain connectivity in early multiple sclerosis". Tur C, Grussu F et al. *Multiple Sclerosis Journal* (2020), 26(7): 774-785, doi: [10.1177/1352458519845105](https://doi.org/10.1177/1352458519845105).
- NeuroImage* 2020b "Cross-scanner and cross-protocol multi-shell diffusion MRI data harmonization: Algorithms and results". Ning L, Bonet-Carne E, Grussu F et al. *NeuroImage* (2020), 221: 117128, doi: [10.1016/j.neuroimage.2020.117128](https://doi.org/10.1016/j.neuroimage.2020.117128).
- NeuroImage* 2020a "Generalised boundary shift integral for longitudinal assessment of spinal cord atrophy". Prados F, ..., Grussu F et al. *NeuroImage* (2020), 209: 116489, doi: [10.1016/j.neuroimage.2019.116489](https://doi.org/10.1016/j.neuroimage.2019.116489).
- Magn Reson Med* 2019 "Fast bound pool fraction mapping via steady-state magnetization transfer saturation using single-shot EPI". Battiston M, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2019), 82: 1025-1040, doi: [10.1002/mrm.27792](https://doi.org/10.1002/mrm.27792).

- NeuroImage 2019 "Cross-scanner and cross-protocol diffusion MRI data harmonisation: a benchmark database and evaluation of algorithms". Tax CMW, Grussu F et al. NeuroImage (2019), 195: 285-299, doi: [10.1016/j.neuroimage.2019.01.077](https://doi.org/10.1016/j.neuroimage.2019.01.077).
- Sci Rep 2018 "Structural cortical network reorganization associated with early conversion to multiple sclerosis". Tur C, ..., Grussu F et al. Scientific Reports (2018), 8: 10715, doi: [10.1038/s41598-018-29017-1](https://doi.org/10.1038/s41598-018-29017-1).
- Magn Reson Med 2018b "An optimized framework for quantitative magnetization transfer imaging of the cervical spinal cord in vivo". Battiston M, Grussu F et al. Magnetic Resonance in Medicine (2018) 79(5): 2576-2588, doi: [10.1002/mrm.26909](https://doi.org/10.1002/mrm.26909).
- Magn Reson Med 2018a "Fast and reproducible in vivo T1 mapping of the human cervical spinal cord". Battiston M, ..., Grussu F et al. Magnetic Resonance in Medicine (2018), 79(4): 2142-2148, doi: [10.1002/mrm.26852](https://doi.org/10.1002/mrm.26852).
- NeuroImage 2017 "Spinal cord grey matter segmentation challenge". Prados F, ..., Grussu F et al. NeuroImage (2017), 152:312-329, doi: [10.1016/j.neuroimage.2017.03.010](https://doi.org/10.1016/j.neuroimage.2017.03.010).
- PlosOne 2016 "Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot in vivo study of the healthy spinal cord at 3T". Yiannakas MC, Grussu F et al. PlosOne (2016), 11(10): e0164890, doi: [10.1371/journal.pone.0164890](https://doi.org/10.1371/journal.pone.0164890).

CONFERENCE PAPERS: FIRST AUTHORSHIP

- CDMRI 2020 "Deep learning model fitting for diffusion-relaxometry: a comparative study". Grussu F, Battiston M, Palombo M, Schneider T, Wheeler-Kingshott CAM, Alexander DC. Proc of 2020 MICCAI Workshop on Computational Diffusion MRI, 2021, 159-172, doi: [10.1007/978-3-030-73018-5_13](https://doi.org/10.1007/978-3-030-73018-5_13). Corresponding author.

CONFERENCE PAPERS: CO-AUTHORSHIP

- MICCAI 2022 "Progressive subsampling for oversampled data - application to quantitative MRI". Blumberg SB, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2022, Lecture Notes in Computer Science, 13436: 421-431, doi: [10.1007/978-3-031-16446-0_40](https://doi.org/10.1007/978-3-031-16446-0_40).
- CDMRI 2019 "Acquiring and predicting multidimensional diffusion (MUDI) data: an open challenge". Pizzolato M, ..., Grussu F et al. Proc of 2019 MICCAI Workshop on Computational Diffusion MRI, 2020, 195-208, doi: [10.1007/978-3-030-52893-5_17](https://doi.org/10.1007/978-3-030-52893-5_17).
- CDMRI 2018b "Multi-shell diffusion MRI harmonisation and enhancement challenge (MUSHAC): progress and results". Ning L, ..., Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 217-224, doi: [10.1007/978-3-030-05831-9_18](https://doi.org/10.1007/978-3-030-05831-9_18).
- CDMRI 2018a "Spatial characterisation of fibre response functions for spherical deconvolution in multiple sclerosis". Tur C, Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 265-279, doi: [10.1007/978-3-030-05831-9_21](https://doi.org/10.1007/978-3-030-05831-9_21).
- MICCAI 2016 "Bayesian image quality transfer". Tanno R, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2016, Lecture Notes in Computer Science, 9901: 265-273, doi: [10.1007/978-3-319-46723-8_31](https://doi.org/10.1007/978-3-319-46723-8_31).

BOOK CHAPTERS

- CRC Press 2018 Chapter 8: "D — the diffusion of water (DTI)". Grussu F and Wheeler-Kingshott CAM. "Quantitative MRI of the brain" (2nd edition, 2018), Cercignani M, Dowell N and Tofts P editors. ISBN 978-1-138-03285-9, doi: [10.1201/b21837](https://doi.org/10.1201/b21837).

BOOK EDITING

- CDMRI 2018 “Computational Diffusion MRI”. Bonet-Carne E, Grussu F, Ning L, Sepehrband F and Tax C editors. Proc. of 2018 MICCAI Workshop on “Computational Diffusion MRI”, Granada, Spain, 20/09/2018. ISBN: 978-3-030-05830-2, doi: 10.1007/978-3-030-05831-9.
- CDMRI 2017 “Computational Diffusion MRI”. Kaden E, Grussu F, Ning L, Tax C and Veraart J editors. Proc. of 2017 MICCAI Workshop on “Computational Diffusion MRI”, Quebec City, Canada, 10/09/2017. ISBN: 978-3-319-73839-0, doi: 10.1007/978-3-319-73839-0.

PRIZES AND AWARDS

- 2023 **Best oral paper award**, 2023 annual meeting of the **Iberian Chapter** of the International Society for Magnetic Resonance in Medicine (ISMRM), Trainee competition for abstract Grussu F et al, Proc of ISMRM Iberian Chapt. 2023.
- 2021 **3rd prize**, 2021 ISMRM **MR of Cancer Study Group**, Trainee competition for abstract Grussu F et al, Proc of ISMRM 2021, p.0699.
- 2021 **Magna cum Laude** award, abstract (Grussu et al, p.0699, ISMRM 2021).
- 2020 **Magna cum Laude** award, abstract (Grussu et al, p.1035, ISMRM 2020).
- 2020 **2nd prize** (shared), 2020 ISMRM British and Irish Chapter “**Mansfield Research Innovation Award**” for abstract Grussu F et al, Proc of ISMRM 2020.
- 2019 **1st prize** in the “Multi-dimensional Diffusion Imaging” (**MUDI**) challenge at 2019 CDMRI MICCAI Workshop (Shenzhen, China, 17/10/2019) (**Team**: Grussu F, Blumberg SB, Ianus A, Mertzaniidou T, Alexander DC; **Method**: SARDU-Net).
- 2018-2020 **Elected trainee representative** for the **White Matter Study Group** of the International Society for Magnetic Resonance in Medicine (ISMRM).
- 2018 & 2019 “**Distinguished reviewer**” Award for Magnetic Resonance in Medicine, awarded at the 2018 and 2019 ISMRM annual meetings.
- 2018 **Magna cum Laude** award, abstract (Grussu et al, p.466, ISMRM 2018).
- 2017 Poster short-listed for presentation at the 2017 ISMRM *Diffusion study group* (Grussu et al, p.3399, ISMRM 2017).
- 2017 **Magna cum Laude** award, abstract (Grussu et al, p.3399, ISMRM 2017).
- 2016 Abstract submission among best 5 in the “*Validation*” session, ISMRM workshop *Breaking the barriers of diffusion MRI*.
- 2016 Poster short-listed for presentation at the 2016 ISMRM *Diffusion study group* (Grussu et al, p.2009, ISMRM 2016).
- 2015 Young Investigators poster competition finalist (80 selected), European Committee for Treatment and Research in Multiple Sclerosis (ECTRIMS) 2015.
- 2015 **Magna cum Laude** award, abstract (Grussu et al, p.0909, ISMRM 2015).
- 2015 **Magna cum Laude** award, abstract (Grussu et al, p.0154, ISMRM 2015).
- 2013 **1st prize**, Master’s degree awards, Mòguru council, Italy (1st prize).
- 2013 Poster short-listed as a finalist of the 2013 ISMRM *White Matter Study Group* poster competition.
- 2012 **2012 Master’s degree thesis prize**, Gruppo Nazionale di Bioingegneria (GNB, National Bioengineering Group).

- 2009 **Award for the best student graduating in Biomedical Engineering (BEng)** in 2009, University of Cagliari, Italy.
- 2008-2011 **"Assegno di Merito" (Merit cheque)** by the Sardinian regional government for excellence in Academic Studies, obtained yearly from 2008 to 2011.

AWARDS IN SUPERVISORY ROLES

- 2024 **Best application (poster)** by the Diffusion Study Group of the International Society for Magnetic Resonance in Medicine (ISMRM) to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Summa cum Laude** award to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Magna cum Laude** award to student Athanasios Grigoriou, for abstract (Grigoriou A et al, and Grussu F, p.0699, ISMRM 2024).

TRAVEL GRANTS

- 2020 UCL Dept. of neuroinflammation travel grant funding attendance at the 2020 ISMRM meeting (GBP 400) – awarded for April 2020, unused due to COVID-19.
- 2016 ISMRM Trainee Stipend for attendance at ISMRM workshop *"Breaking the barriers of Diffusion MRI (waived registration fees)"*.
- 2015, 2016 UCL School of Life and Medical Sciences Travel Grant funding the attendance at the 2015 (GBP 800) and 2016 (GBP 940) ISMRM meetings.
- 2015 ECTRIMS Travel Grant funding the attendance at the 2015 ECTRIMS meeting in Barcelona (Spain) (EUR 400).
- 2013, 2015, 2017 Guarantors of Brain Travel Grant funding the attendance at the 2013 (GBP 800), 2015 (GBP 500) and 2017 (GBP 800) ISMRM meetings.
- 2012-2015 ISMRM Trainee Stipend supporting attendance at the annual meeting (yearly).

INVITED ORAL COMMUNICATIONS

- BSC 2025 *"Histology-informed computer simulations enhance Magnetic Resonance Imaging (MRI) virtual biopsies in cancer"*. **"Severo Ochoa Research Seminar"**, Barcelona Supercomputing Center (BSC), Barcelona, Spain, 13/11/2025.
- ODELIA Summer School 2025 *"An overview of Magnetic Resonance Imaging (MRI): from basic principles to cutting-edge applications"*. ODELIA consortium on open source swarm learning to empower medical AI. **"2025 ODELIA Summer School"**, Vall d'Hebron Institute of Oncology (VHIO), Barcelona, Spain, 18/09/2025.
- ISMRM Iberian Chapter 2025 *"Quantitative imaging and AI Automation"*. International Society for Magnetic Resonance in Medicine (ISMRM), 2025 annual meeting of the Iberian Chapter. **"2025 ISMRM Iberian Chapter pre-conference Summer School"**, Institut de Bioenginyeria de Catalunya (IBEC), Barcelona (Spain), 02/07/2025.
- ISMRM Workshop 2025 *"Challenges in body diffusion and how to overcome them"*. International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on: **"40 years of Diffusion: past, present and future perspectives"**, Kyoto University, Kyoto (Japan), 18/02/2025.
- EPFL 2024 *"Diffusion MR signal modelling for oncological body imaging"*. **Visitors Talk**, Centre for Biomedical Imaging (CIBM), École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland), 09/10/2024.

ESMRMB workshop 2024	"Modelling body microstructure". ESMRMB 2024 pre-congress workshop " Leaps in Microstructure Imaging: Exploring New Horizons ", Barcelona (Spain), 02/10/2024.
DiPy workshop 2024	"Advanced body diffusion MRI for oncological applications". 2024 DiPy workshop , online, 14/03/2024.
ESMRMB workshop 2023	"Histology-informed body diffusion MRI in oncological applications". ESMRMB 2023 pre-congress workshop " Frontiers in preclinical MRI ", Basel (Switzerland), 04/10/2023.
ESMRMB lectures 2023	"Diffusion MRI in the body". ESMRMB Lectures on MR , "Introduction to diffusion-weighted MR imaging and spectroscopy", Cardiff (UK), 07/09/2023.
DSG ISMRM 2023	"Emerging models in oncology". " Virtual biopsies by diffusion MRI: are we there yet in oncology? ", Virtual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM) Diffusion Study Group (DSG). Online, 19/04/2023.
UCL 2022b	"Data-driven, model-free, deep learning approach for quantitative MRI protocol design". " Microstructure Imaging Meets Machine Learning " (MIML) workshop, University College London, London (UK), 13/05/2022.
ISMRM 2022	"Modeling diffusion in cancer and body". Educational session: "Diffusion" , 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK), 07/05/2022.
University of Verona 2019	" Diffusion MRI data harmonisation ". 2019 School on Brain Connectomics, University of Verona (Italy), 24/09/2019.
ISMRM Italian Chapter 2018	" Axonal dispersion from diffusion MRI: a new marker of microstructural damage ". Italian Association for Magnetic Res. in Medicine, Padua (Italy), 10/05/2018.

MODERATION OF SCIENTIFIC SESSIONS

ECR 2026	Moderator of European Society of Radiology (ESR) Research Committee Session 10 " Real-world applications of imaging biomarkers: from hypothesis to clinical practice ", 05/03/2026, 2026 European Conference of Radiology (ECR), organised by the ESR in Vienna (Austria).
ISMRM Iberian Chapter 2025	Moderator of oral scientific session " Clinical session ", 03/07/2025, 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM), Barcelona (Spain).
ISMRM 2022	Moderator of oral "power pitch" scientific session " Motion correction ", 10/05/2022, 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK).
ISMRM 2021b	Moderator of oral scientific session " Microstructure: Modelling Gray & White Matter Diffusion ", 19/05/2021, 2021 virtual annual meeting of the ISMRM.
ISMRM 2021a	Facilitator of poster session " Diffusion Applications: Brain & Spine ", 18/05/2021, 2021 virtual annual meeting of the ISMRM.

PARTICIPATION IN EVALUATION PANELS

Research Council of Finland 2025	Grant reviewer for the Research Council of Finland, 2025 Call for Clinical Research (online evaluation).
EPFL 2024	PhD defense at the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland). Thesis title: " Lymph node microstructure imaging using diffusion MRI: From simulations to acquisitions ". Exam date: 08/10/2024 (in

person in Lausanne). Role: external examiner. Degree: PhD. Department: Center for Biomedical Imaging (CIBM), EPFL. Thesis link [here](#).

ORGANISATION OF SCIENTIFIC EVENTS

<i>UB Seminar 2026</i>	Institute of Neuroscience Seminar at <i>Universitat de Barcelona</i> 2026: “Quantifying blood-brain barrier permeability using MRI”, 24/04/2026. Organisers and moderators: Roser Sala-Llonch, Francesco Grussu.
<i>ISMRM Iberian Chapter 2025</i>	ISMRM Iberian Chapter: 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM) (Barcelona, scheduled for 3-4/07/2025). Organisers: Marco Rius I (IBEC), Ortega Machuca MA (IBEC), Candiota AP (UAB), Julià-Sapé M (UAB), Muñoz Moreno E (IDIBAPS), Lope-Piedrafita S (UAB), Barba Vert I (UVic), Benito M (UCM), Grussu F (VHIO), Gómez Cabeza D (IBEC).
<i>MIS ISMRM 2021</i>	Member Initiated Symposium at ISMRM 2020: “Looking Beyond Axons: Imaging the Immune System in White Matter”, 19/05/2021. Organisers: Cohen-Adad J, Grussu F, Kolind S.
<i>WMSG ISMRM 2019</i>	ISMRM White Matter Study Group Virtual meeting: “Myelin Imaging in the Spinal Cord at High Field”, 27/06/2019, joint meeting with the High-field Study Group. Organiser: Grussu F. Chair: Cohen-Adad J.
<i>MIS ISMRM 2019</i>	Member Initiated Symposium at ISMRM 2019: “Completing the Circle: Moving Multi-Parametric Neuro MRI into Clinical Practice and Trials”, 15/05/2019. Organisers: Vrenken H, Cohen-Adad J, Grussu F.
<i>CDMRI 2018</i>	MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2018 (Granada, Spain, 20/09/2018). Organisers: Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).
<i>MUSHAC Challenge 2018</i>	MICCAI Challenge: Multi-shell dMRI harmonisation and enhancement (MUSHAC, part of CDMRI 2018). Organisers: Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).
<i>CDMRI 2017</i>	MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2017 (Quebec City, Canada, 10/09/2017). Organisers: Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).
<i>Data Harmonisation Challenge 2017</i>	MICCAI Challenge: Diffusion MRI data harmonisation (part of CDMRI 2017). Organisers: Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).
<i>Spinal Cord Grey Matter Segmentation Challenge 2016</i>	Challenge: Grey Matter Segmentation: What’s there and What’s next? (part of SC MRI Workshop 2016). Organisers: Cohen-Adad J, Prados F, Landman B, Wheeler-Kingshott C, Summers P, Dupont S, Yiannakas M, Smith S, Gergely D, DeLeener B, Grussu F.

SCIENTIFIC PEER REVIEWING

<i>2016-present</i>	Reviewer for Bipolar Disorders; Functional Neurology; Frontiers; Human Brain Mapping; Journal of Magnetic Resonance Imaging; Journal of Neuroscience Methods; Magnetic Resonance in Medicine; NeuroImage; Medical Image Analysis; IEEE Transactions on Medical Imaging; npj Precision Oncology; NMR in Biomedicine; Scientific Data; 2019, 2021, 2022 ISMRM annual meetings; 2021 ISMRM Iberian Chapter post-grad meeting; 2021, 2025 Iberian Chapter annual meeting; 2019, 2020 MICCAI workshop on Computational Diffusion MRI.
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OFFICIAL STUDENT SUPERVISION

<i>MSc, VHIO and UAB 2025/26</i>	Project co-supervision: “Development and application of computational Magnetic
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Resonance Imaging for detecting and monitoring of aggressive cancers". Degree: Master's in "Bioninformatics", Universitat Autònoma de Barcelona (UAB; Barcelona, Spain). Project being carried out at VHIO, Barcelona (Spain). Supervised in: 2025/2026 (ongoing).

PhD, VHIO and
UB 2023-27

Project co-supervision: *"Novel Magnetic Resonance Imaging biomarkers for precision medicine in oncology: integrating multi-omics and real-world data strategies". Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 2023-2027 (ongoing).*

PhD, VHIO and
UB 2022-26

Project co-supervision: *"Histology-informed diffusion MRI simulations and artificial intelligence for cancer microstructure characterization". Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at the Vall d'Hebron Institute of Oncology (VHIO), Barcelona (Spain). Supervised in: 2022-2026 (ongoing).*

PhD, VHIO and
UB 2022-26

Project co-supervision: *"A simulation framework for characterizing cancer microvasculature with histology-informed diffusion MRI". Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 30/09/2022-23/04/2026 (completed). Final mark: excellent and cum laude.*

MEng, VHIO and
TU Delft 2022/23

Project co-supervision: *"Unraveling tumour microstructure through diffusion MRI using histology-powered artificial intelligence". Degree: Master's in "Biomedical Engineering", Delft University of Technology (TU Delft), (Delft, The Netherlands). Project carried out at VHIO, Barcelona (Spain). Supervised in: 09-2022/06-2023 (completed).*

MEng, VHIO and
UI La Rioja
2022/23

Project co-supervision: *"Prediction of advanced biomarkers from clinical diffusion Magnetic Resonance Imaging". Degree: Master's in "Artificial Intelligence", Universidad Internacional de La Rioja (Logroño, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 2022/2023 (completed).*

MRes, UCL
2018/19

Project co-supervision: *"Improving the differential diagnosis between Neuromyelitis Optica Spectrum Disorder and Multiple Sclerosis using MRI". Degree: MRes in "Clinical Neuroscience", Institute of Neurology, University College London (UCL, London, UK). Supervised in: 2018/2019 (completed).*

MSc, UCL
2018/19

Project co-supervision: *"Evaluation of quantitative MRI indices reproducibility across scanner upgrade". Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2018/2019 (completed).*

MSc, UCL
2017/18

Project co-supervision: *"Investigation of multi-component T₁ relaxation at 3 Tesla". Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2017/2018 (completed).*

MSc, UCL
2015/16

Project co-supervision: *"Evaluation of strategies for co-registration between quantitative and anatomical magnetic resonance images of the human spinal cord". Degree: MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). Supervised in: 2015/2016 (completed).*

TEACHING EXPERIENCE

UB Lecture

Lecture: *"Advanced medical imaging: a journey through engineering, physics and computer science". Subject/module: "Artificial intelligence in biomedical engineering". Degree: BEng in "Biomedical engineering", Universitat de Barcelona (Barcelona, Spain). Conveyed: 12/05/2026. Face-to-face teaching: 2 hours.*

UVi-UCC Lecture
and Workshop

Lecture and Workshop: *"Medical image analysis". Subject/module: "Interactomics". Degree: MSc in "Omics Data Analysis", Universitat de*

Vic-Universitat Central de Catalunya, UVic-UCC (Vic and Barcelona, Spain).

Conveyed: 17/03/2026. **Face-to-face teaching:** 4 hours.

UCL Lecture	Lecture: “Image optimisation: SNR, CNR and sources of artifacts”. Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2019/20, 2017/18, 2016/17, 2015/16. Face-to-face teaching: 4 hours (1 hour per year). Additional tasks: exam preparation and assessment.
UCL Workshop	Workshop: Hands-on with a portable MRI scanner . Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2017/18, 2016/17, 2015/16. Face-to-face teaching: 12 hours (4 hour per year).
UniPV Workshop	Workshop: “Model fitting for quantitative MRI”. Degree: MEng in “Biomedical Engineering”, University of Pavia (Pavia, Italy). Conveyed: 2016/17. Face-to-face teaching: 2 hours. Additional tasks: exam preparation and assessment.
UCL Lecture	Lecture: “Magnetic resonance image formation”. Degree: MSc in “Advanced biomedical imaging”, Centre for Advanced Biomedical Imaging, UCL (London, UK). Conveyed: 2015/16. Face-to-face teaching: 1 hour.

OPEN SCIENCE

Code released through GitHub (<http://fragrussu.github.io> and <http://github.com/radiomicsgroup>)

GitHub 2024c	Histo-μSim: histology-informed cancer diffusion MRI (link).
GitHub 2024b	SpinFlowSim: diffusion MRI simulator in vascular networks (link).
GitHub 2024a	BodyMRITools: python code for body diffusion MRI processing (link).
GitHub 2022	MChepato: Code and synthetic data for Grussu et al, MRM 2022 (link). Available in Zenodo as record 6645258, doi: 10.5281/zenodo.6645258 .
GitHub 2020b	qMRI-Net: MRI signal model fitting based on artificial intelligence (link).
GitHub 2020a	SARDU-Net: data-driven, model-free quantitative MRI protocol design (link).
GitHub 2019b	MyRelax: tools for myelin and relaxation MRI analyses (link to the latest version). Version 1.0.0 available in Zenodo as record 4561898, doi: 10.5281/zenodo.4561898 .
GitHub 2019a	MRITools: tools for handling and managing research MRI scans (link).
GitHub 2016	StructureTensorToolbox: tools for analysis of 2D histological images (link).

Data sets released through Zenodo

Zenodo 2024b	“Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo- μ Sim): ex vivo mouse data”. Grussu F, Grigoriou A, Macarro C and Perez-Lopez R. Data set. Zenodo record 14559355, 2024, doi: 10.5281/zenodo.14559355 .
Zenodo 2024a	“Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo- μ Sim): histology substrates”. Grigoriou A, Macarro C, Perez-Lopez R and Grussu F. Data set. Zenodo record 14559103, 2024, doi: 10.5281/zenodo.14559103 .

DISSEMINATION TO THE WIDER PUBLIC

2024b	Lab demonstration: “Uso de la inteligencia artificial en la investigación contra el
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cáncer” (“Using artificial intelligence in cancer research”) at the 2024 European Researchers’ Night (27/09/2024), CosmoCaixa, Barcelona (Spain).

- 2024a Video: “Resonancia magnética para combatir el cáncer” (“Magnetic Resonance Imaging to fight cancer”), released through the YouTube channel of the Vall d’Hebron Institute of Oncology (Barcelona, Spain) ([link](#), video in Spanish).
- 2022b Science fair: “Demostración de análisis avanzado de imágenes por resonancia magnética y datos co-localizados de microscopía” (“Demonstration of advanced analysis of MR images and co-localised microscopy”) at the 2022 European Researchers’ Night (30/09/2022), CosmoCaixa, Barcelona (Spain).
- 2022a “Innovating Magnetic Resonance Imaging to fight diseases”. Università della Terza Età (UniTre), Mòguru, Italy (12/04/2021). Talk given to a general audience in Sardinian language.
- 2017 “Why to get vaccinated and avoid misinformation: the scientific method in modern medicine”. Event for a general audience in Italian and Sardinian, Mòguru, Italy (17/08/2017). Organisers: Grussu F, Tur C, Coccollone E, Broccia S.
- 2015 Participation at MS Frontiers 2015 (29-30/06/2015), organised by the UK Multiple Sclerosis Society and bringing together researchers and Multiple Sclerosis patients.
- 2013 UCL stall at *Science Uncovered*, 2013 European Researcher’s Night (28/09/2013), London Science Museum.

CONFERENCE & WORKSHOP PROCEEDINGS: FIRST AUTHORSHIP

- ECR 2026 “Response assessment and prognosis prediction in cancer immunotherapy using multi-parametric MRI: preliminary evidence”. Grussu F et al. European Conference of Radiology (ECR) 2026, C-16681 (EPOS™oral + digital poster presentation, 07/03/2026), doi: [10.26044/ecr2026/C-16681](https://doi.org/10.26044/ecr2026/C-16681).
- ESMRMB 2024 “Histological interpretation of Susceptibility-Perturbation MRI in human tumours of the liver”. Grussu F et al. European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2024 (traditional poster presentation).
- ISMRM 2024b “Histology-informed biophysical diffusion MRI model selection for enhanced liver cancer immunotherapy assessment”. Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2024 (**oral presentation**).
- ISMRM 2024a “Two-axon population (TAP) modelling for large axon diffusion imaging in the peripheral nervous system”. Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2024 (**oral presentation**).
- ISMRM Iberian 2023 “Extra-cellular liver diffusion modelling at high b-value: a preclinical MRI-histology study”. Grussu F et al. Iberian Chapter of the ISMRM 2023 (**oral presentation**).
- ISMRM 2022b “Histological correlates of DR-HIGADOS microstructural metrics in the mouse and human liver”. Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2022 (power-pitch presentation).
- ISMRM 2022a “Inter-scanner reproducibility and variability assessment of advanced liver diffusion MRI metrics”. Grussu F et al. ISMRM 2022 (d-poster presentation).
- ISMRM 2021b “DR-HIGADOS: a new diffusion-relaxation framework for clinically feasible microstructural imaging of the liver”. Grussu F et al. ISMRM 2021 (**oral presentation, Magna cum Laude award**).
- ISMRM 2021a “Investigating the relationship between diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time”. Grussu F et al. International Society for Magnetic Resonance in Medicine (ISMRM) 2021 (**d-poster presentation**).

ISMRM 2020	<i>"SARDU-Net: a new method for model-free, data-driven experiment design in quantitative MRI"</i> . Grussu F et al. ISMRM 2020 (power-pitch presentation, Magna cum Laude award).
ISMRM 2019	<i>"Clinically viable g-ratio imaging with unified readout at 3T: evaluation and comparison"</i> . Grussu F et al. ISMRM 2019 (e-poster presentation).
ISMRM 2018b	<i>"Magnitude versus complex-valued images for spinal cord diffusion MRI: which one is best?"</i> . Grussu F et al. ISMRM 2018 (oral presentation, Magna cum Laude award).
ISMRM 2018a	<i>"A unified signal readout improves denoising of multi-modal spinal cord MRI"</i> . Grussu F et al. ISMRM 2018 (poster presentation).
ISMRM 2017b	<i>"Origin of the time dependence of the diffusion-weighted signal in spinal cord white matter"</i> . Grussu F et al. ISMRM 2017 (oral presentation).
ISMRM 2017a	<i>"A unified signal readout for reproducible multimodal characterisation of brain microstructure"</i> . Grussu F et al. ISMRM 2017 (e-poster presentation, Magna cum Laude award, finalist at the Diffusion Study Group competition).
Brain School 2017	<i>"Whole-brain macromolecular tissue volume mapping: A comparison of imaging readouts at 3 Tesla"</i> . Grussu F et al. School of Brain Cells and Circuits "Camillo Golgi". Frontiers ISBN 978-288945-584-3 (poster presentation).
ISMRM Scientific Workshop 2016	<i>"Optimal histological quantification of neurite orientation dispersion for the validation of diffusion MRI"</i> . ISMRM Scientific workshop "Breaking the barriers of diffusion MRI" (poster presentation + Power Pitch).
ISMRM 2016	<i>"Axon diameter distribution influences diffusion-derived axonal density estimation in the human spinal cord: in silico and in vivo evidence"</i> . Grussu F et al. ISMRM 2016 (poster presentation, finalist at the Diffusion Study Group competition).
ECTRIMS 2015	<i>"Quantitative histological validation of NODDI MRI indices of neurite morphology in multiple sclerosis spinal cord"</i> . Grussu F et al. European Committee for Research and Treatment of Multiple Sclerosis (ECTRIMS) 2015 (poster presentation, short-listed for poster prize competition).
MS Frontiers 2015	<i>"Histological correlates of NODDI in the multiple sclerosis spinal cord"</i> . Grussu F et al. MS Frontiers 2015, annual scientific meeting of the UK Multiple Sclerosis Society (oral and poster presentation).
ISMRM 2015b	<i>"Quantitative histological correlates of NODDI orientation dispersion estimates in the human spinal cord"</i> . Grussu F et al. ISMRM 2015 (oral presentation, Magna cum Laude award).
ISMRM 2015a	<i>"Histological metrics confirm microstructural characteristics of NODDI indices in multiple sclerosis spinal cord"</i> . Grussu F et al. ISMRM 2015 (oral presentation, Magna cum Laude award).
British Chapter of the ISMRM 2014	<i>"Characterisation of single-shell NODDI fitting in spinal cord grey and white matter"</i> . Grussu F et al. British Chapter of the ISMRM 2014 (poster presentation).
ISMRM 2014b	<i>"Neurite orientation dispersion and density imaging of the cervical cord in vivo"</i> . Grussu F et al. ISMRM 2014 (poster presentation).
ISMRM 2014a	<i>"Single-shell diffusion MRI NODDI with in vivo cervical cord data"</i> . Grussu F et al. ISMRM 2014 (poster presentation).
ISMRM Workshop 2013	<i>"In vivo estimation of neuronal orientation dispersion and density of the human spinal cord"</i> . ISMRM Scientific workshop "Multiple sclerosis as a whole-brain disease" (oral presentation).
ISMRM 2013	<i>"Towards spinal cord microstructure mapping with the neurite orientation dispersion</i>

and density imaging". Grussu F et al. ISMRM 2013 (poster presentation, finalist at the White Matter Study Group poster competition).

CONFERENCE & WORKSHOP PROCEEDINGS: SENIOR AUTHORSHIP

- ISMRM Iberian Chapter 2025 *"Simplicity is a virtue: histology-informed model comparison selects simple diffusion representations in a colorectal cancer metastasis specimen"*. Grigoriou A, ..., and Grussu F. 2025 ISMRM Iberian Chapter annual meeting, p.49 (poster presentation). **2nd prize for the best poster presentation.**
- ISMRM Workshop 2025b *"Which Microvascular Properties Can We Probe in Clinical Settings with Diffusion MRI?"*. Voronova A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (oral presentation).
- ISMRM Workshop 2025b *"Histology-Informed Microstructural Diffusion Simulations (Histo- μ Sim) for Enhanced Diffusion MRI Parameter Estimation in Cancer"*. Grigoriou A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (power-pitch presentation), p. 202.
- ESMRMB 2024b *"Biologically-realistic blood flow simulations reveal complex features of vascular IVIM signals"*. Voronova A, ..., and Grussu F. ESMRMB 2024 (traditional poster presentation).
- ESMRMB 2024a *"Histology-informed cell size distribution mapping with diffusion MRI"*. Grigoriou A, ..., and Grussu F. ESMRMB 2024 (traditional poster presentation).
- ISMRM 2024b *"FlowSim: a blood flow simulator for histology-informed diffusion MRI micro-vasculature mapping in cancer"*. Voronova A, ..., and Grussu F. ISMRM 2024 (power-pitch presentation, **Summa cum Laude** award, **Diffusion Study Group prize, best application (poster category)**).
- ISMRM 2024a *"A Monte Carlo simulation framework for histology-informed diffusion MRI parameter estimation in cancer"*. Grigoriou A, ..., and Grussu F. ISMRM 2024 (power-pitch presentation, **Magna cum Laude** award).
- ISMRM Scientific Workshop 2022 *"A systematic comparison of machine learning approaches for diffusion-relaxation MRI protocol enhancement in advanced solid tumours"*. Macarro C, ..., and Grussu F. ISMRM Workshop on Diffusion MRI From Research to Clinic 2022 (poster presentation).
- ISMRM 2019 *"Sensitivity of NODDI and two-compartment SMT parameter maps in multiple sclerosis"*. Johnson D, ..., and Grussu F. ISMRM 2019 (e-poster presentation).

CONFERENCE & WORKSHOP PROCEEDINGS: SELECTED CO-AUTHORSHIP

- ISMRM 2023 *"Decomposition of clinical ADC into intracellular and extracellular-extravascular contributions in prostate cancer using histology"*. Garcia-Ruiz A et al. ISMRM 2023 (oral presentation).
- ENA Symposium 2022 *"Non-invasive biomarkers for response and survival prediction in patients with advanced solid tumours treated with immune checkpoint inhibitors (ICIs)"*. Bernatowicz K et al. European Journal of Cancer 174S1 (2022) S3–S128, doi: [10.1016/S0959-8049\(22\)00988-1](https://doi.org/10.1016/S0959-8049(22)00988-1) (EORTC-NCI-AACR (ENA) Symposium, 2022 October 26-28, Barcelona, Spain; poster presentation).
- ISMRM Workshop 2022 *"Decoding liver intra-tumour heterogeneity with co-localized CT and multi-parametric MRI"*. Prior Palomares O, Grussu F, et al. ISMRM Workshop on Diffusion MRI From Research to Clinic 2022 (oral presentation).
- ISMRM 2022b *"Deep-learning-informed parameter estimation improves reliability of spinal cord diffusion MRI"*. Gong T et al. ISMRM 2022 (oral presentation).

ISMRM 2022a	<i>"Deep learning voxelwise classification of primary central nervous system lymphoma using DSC-PWI normalized time-intensity curves". Garcia-Ruiz A et al. ISMRM 2022 (d-poster presentation).</i>
ISMRM 2021	<i>"Associations between cervical cord sodium concentration, neuronal density and macromolecular tissue volume in spinal cord injury". Solanky B et al. ISMRM 2021 (oral presentation).</i>
ISMRM 2020	<i>"Acquiring and predicting Multi-dimensional Diffusion (MUDI) data: an open challenge". Pizzolato M et al. ISMRM 2020 (oral presentation).</i>
ISMRM 2020b	<i>"New potential MRI markers of glial scarring and tissue damage in multiple sclerosis spinal cord pathology using diffusion MRI". Palombo M et al. ISMRM 2020 (power-pitch presentation).</i>
ISMRM 2020a	<i>"SENSE reconstruction with simultaneous 2D phase correction and channel-wise noise removal (SPECTRE)". Powell E et al. ISMRM 2020 (d-poster presentation).</i>
ISMRM 2020	<i>"Quantitative MRI of the spinal cord: reproducibility and normative values across 40 sites". Alonso-Ortiz E L et al. ISMRM 2020 (oral presentation).</i>
ISMRM 2019b	<i>"Cross-scanner and cross-protocol harmonisation of multi-shell diffusion MRI data: open challenge and evaluation results". Ning L et al. ISMRM 2019 (oral presentation).</i>
ISMRM 2019a	<i>"Bound Pool Fraction mapping via steady-state MT saturation using single-shot EPI". Battiston M et al. ISMRM 2019 (oral presentation).</i>
ISMRM 2018b	<i>"Cross-vendor and cross-protocol harmonisation of diffusion MRI data: a comparative study". Tax C et al. ISMRM 2018 (oral presentation).</i>
ISMRM 2018a	<i>"Consensus acquisition protocol for quantitative MRI of the cervical spinal cord at 3T". Alley S et al. ISMRM 2018 (oral presentation).</i>
ECTRIMS 2017c	<i>"Application of Neurite Orientation Dispersion and Density Imaging (NODDI) in clinically isolated syndrome (CIS)". Collorone S et al. ECTRIMS 2017 (poster presentation).</i>
ISMRM 2017b	<i>"Boundary shift integral to compute brain and cervical spinal cord longitudinal atrophy on the same 3D T1 brain images in multiple sclerosis". Prados F et al. ISMRM 2017 (oral presentation).</i>
ISMRM 2017a	<i>"Impact of acquisition strategies and spherical deconvolution algorithms on brain connectivity mapping in early multiple sclerosis". Tur C et al. ISMRM 2017 (oral presentation).</i>
ECTRIMS 2016b	<i>"Computing spinal cord atrophy using the boundary shift integral: a more powerful outcome measure for clinical trials?". Prados F et al. ECTRIMS 2016 (poster presentation).</i>
ECTRIMS 2016a	<i>"Neurite orientation dispersion and density imaging (NODDI) reflects early microstructural brain tissue changes in clinically isolated syndrome (CIS)". Collorone S et al. ECTRIMS 2016 (poster presentation).</i>
ISMRM 2016b	<i>"Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot in vivo study of the healthy spinal cord using a clinical 3T MR system". Yiannakas M et al. ISMRM 2016 (e-poster presentation).</i>
ISMRM 2016a	<i>"Atrophy computation in the spinal cord using the boundary shift integral". Prados F et al. ISMRM 2016 (oral presentation).</i>
AAN 2016b	<i>"No Differences in spinal cord white and grey matter diffusion abnormalities between neuromyelitis optica spectrum disorder and multiple sclerosis". Cortese R et al. American Accademy of Neurology (AAN) 2016 (poster presentation).</i>

AAN 2016a	<i>"Neurite orientation dispersion and density imaging (NODDI) at the onset of clinically isolated syndrome (CIS): new insights in the early microstructural brain tissue changes". Collorone S et al. AAN 2016 (dual presentation).</i>
ISMRM 2015	<i>"Combined sodium-NODDI: towards quantitative in vivo intracellular and intraneurite sodium measures at 3T". Solanky B et al. ISMRM 2015 (e-poster presentation).</i>
ISMRM 2014	<i>"An investigation of brain neurite density and dispersion in multiple sclerosis using single shell diffusion imaging". Magnollay L et al. ISMRM 2014 (poster presentation).</i>
ECTRIMS 2013	<i>"Application of neurite orientation dispersion and density imaging (NODDI) to relapsing remitting multiple sclerosis (RRMS)". Magnollay L et al. ECTRIMS 2013 (poster presentation).</i>
ECTRIMS 2013	<i>"Neurite orientation dispersion and density imaging in the multiple sclerosis spinal cord". Kearney H et al. ECTRIMS 2013 (e-poster presentation).</i>
NCM 2012	<i>"Algorithms for shaping the dynamics of a bidirectional neural interface". Semprini M et al. Society for the Neural Control of Movement (NCM) 2012 (poster presentation).</i>

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